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2015.0 RANGE ROVER (LG), 303-12

INTAKE AIR DISTRIBUTION AND FILTERING - TDV8 4.4L DIESEL

PRINCIPLES OF OPERATION

For a detailed description of the intake air distribution and filtering system and operation, refer to the relevant Description and Operation section in the Workshop Manual.

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Hoses and ducts: condition and installation ▪ Air cleaner element condition and installation ▪ Restricted air intake ▪ Vacuum hoses condition and installation ▪ Pipework to/from turbochargers: condition and installation ▪ Turbochargers: condition and installation ▪ Charge air coolers	▪ Fuse(s) ▪ Wiring harness(es) ▪ Loose or corroded electrical connector(s) ▪ Mass air flow (MAF) sensors ▪ Manifold absolute pressure/temperature (MAPT) sensors ▪ Intake air temperature (IAT) sensor ▪ The IAT sensor is built into the right-hand MAF sensor

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Vehicle does not start/hard starting	▪ Restricted /Blocked air intake ▪ Restricted /Blocked air cleaner element	Check the intake air system for blockages or restriction. Rectify as necessary.
Poor performance	▪ Intake air system fault ▪ Turbocharger fault(s) ▪ Exhaust gas recirculation (EGR) valve (s) fault ▪ Low fuel pressure ▪ Restricted exhaust system	Check the intake air system for blockages or restriction. Rectify as necessary. Check for DTCs indicating a turbocharger, EGR valve or fuel pressure fault. Rectify as necessary. Check the exhaust system for evidence of damage or restriction. Rectify as necessary.
Excessive intake noise	▪ Intake air leak after the turbocharger ▪ Intake pipe disconnected /damaged after the air cleaner ▪ Air cleaner assembly incorrectly assembled /damaged	Check the intake air system for loose or disconnected hoses or ducts. Check the hoses and ducts for damage, splits, etc. Rectify as necessary.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - 3.6L (TdV8) Diesel, DTC: Module Name: Engine Control Module (100-00, Description and Operation).